

Assignment: 14

PROBLEM STATEMENT: Create an Elastic IP for an Instance.

STEP 1-> create an Instances.

Launch an instance | EC2 | ap-south-1 | pxeh2oy622al5wws7konnjqu... | ydr567gjtbnxig6yjcjon5ja0uc... | +

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name: panins14 [Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Search our full catalog including 1000s of application and OS images

Recents Quick Start

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

Browse more AMIs
Including AMIs from AWS, Marketplace and the Community

Summary

Number of instances [Info](#): 1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-0e35ddb05955cf57

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of

[Cancel](#) [Launch instance](#) [Preview code](#)

Launch an instance | EC2 | ap-south-1 | pxeh2oy622al5wws7konnjqu... | ydr567gjtbnxig6yjcjon5ja0uc... | +

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:

Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-0e35ddb05955cf57

Architecture **AMI ID** **Publish Date** **Username** [Info](#)

64-bit (x86) ami-0e35ddb05955cf57 2025-03-05 ubuntu [Verified provider](#)

Instance type [Info](#) [Get advice](#)

Instance type

t2.micro [Free tier eligible](#)

Family: t2 1 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0124 USD per Hour On-Demand Windows base pricing: 0.017 USD per Hour
On-Demand RHEL base pricing: 0.0268 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0142 USD per Hour
On-Demand SUSE base pricing: 0.0124 USD per Hour

[Additional costs apply for AMIs with pre-installed software](#)

Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

spkey1 [Create new key pair](#)

Summary

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[Cancel](#) [Launch instance](#) [Preview code](#)

Launch an instance | EC2 | ap-south-1 | console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:

Enable
Additional charges apply when outside of free tier allowance

Firewall (security groups) | Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-3' with the following rules:

- ☒ Allow SSH traffic from
Helps you connect to your instance
Anywhere
0.0.0.0/0
- ☒ Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server
- ☒ Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Configure storage | Info | Advanced
1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Summary
Number of instances | Info
1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...read more
ami-0e35ddab05955cf57

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of

Cancel Launch instance Preview code

Launch an instance | EC2 | ap-south-1 | console.aws.amazon.com/ec2/home?region=ap-south-1#LaunchInstances:

Success
Successfully initiated launch of instance (i-0a84a91aa71762b54)

Launch log

Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing and free tier usage alerts
To manage costs and avoid surprise bills, set up email notifications for billing and free tier usage thresholds.
Create billing alerts

Connect to your instance
Once your instance is running, log into it from your local computer.
Connect to instance
Learn more

Connect an RDS database
Configure the connection between an EC2 instance and a database to allow traffic flow between them.
Connect an RDS database
Create a new RDS database
Learn more

Create EBS snapshot policy
Create a policy that automates the creation, retention, and deletion of EBS snapshots.
Create EBS snapshot policy

STEP 2-> Go to instance and save its Public IPv4 address in a text file for future comparison.

Instance details | EC2 | ap-south-1

pxeh2oy622al5wsws7komrnjqun x ydr567igtbnxig6ycjion5ja0uc x +

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#InstanceDetails:instanceId=i-0a84a91aa71762b54

EC2 > Instances > i-0a84a91aa71762b54

Instance summary for i-0a84a91aa71762b54

Updated less than a minute ago

[Public IPv4 address copied](#)

Instance ID
i-0a84a91aa71762b54

IPv6 address
-

Hostname type
IP name: ip-172-31-15-41.ap-south-1.compute.internal

Answer private resource DNS name
IPv4 (A)
-

Auto-assigned IP address
3.7.65.189 [Public IP]

IAM Role
-

IMDSv2
Required

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-15-41.ap-south-1.compute.internal

Instance type
t2.micro

VPC ID
vpc-055d721b1004494c1

Subnet ID
subnet-067810c72322625a8

Instance ARN
arn:aws:ec2:ap-south-1:676206917408:instance/i-0a84a91aa71762b54

Private IPv4 addresses
172.31.15.41

Public IPv4 DNS
ec2-3-7-65-189.ap-south-1.compute.amazonaws.com | [open address](#)

Elastic IP addresses
-

AWS Compute Optimizer finding
[Opt-in to AWS Compute Optimizer for recommendation](#)
[Learn more](#)

Auto Scaling Group name
-

Managed
false

[Connect](#) [Instance state](#) [Actions](#)

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STEP 3-> With the instance selected, open the Instance State dropdown menu and select "Stop Instance" option to temporarily put the instance in the sleep state. Now again start this instance.

Instance details | EC2 | ap-south-1

pxeh2oy622al5wsws7komrnjqun x ydr567igtbnxig6ycjion5ja0uc x +

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#InstanceDetails:instanceId=i-0a84a91aa71762b54

EC2 > Instances > i-0a84a91aa71762b54

Successfully initiated stopping of i-0a84a91aa71762b54

Instance summary for i-0a84a91aa71762b54 (panins14)

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Instance ID
i-0a84a91aa71762b54

IPv6 address
-

Hostname type
IP name: ip-172-31-15-41.ap-south-1.compute.internal

Answer private resource DNS name
IPv4 (A)
-

Auto-assigned IP address
-

IAM Role
-

IMDSv2
-

Public IPv4 address
-

Instance state
Stopped

Private IP DNS name (IPv4 only)
ip-172-31-15-41.ap-south-1.compute.internal

Instance type
t2.micro

VPC ID
vpc-055d721b1004494c1

Subnet ID
subnet-067810c72322625a8

Instance ARN
arn:aws:ec2:ap-south-1:676206917408:instance/i-0a84a91aa71762b54

Private IPv4 addresses
172.31.15.41

Public IPv4 DNS
-

Elastic IP addresses
-

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Auto Scaling Group name
-

Managed
-

[Connect](#) [Instance state](#) [Actions](#)

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Instance details | EC2 | ap-south-1 | pxeh2oy622al5wsw7konnjqu | ydr567igtbnxig6ycjion5ja0uc | +

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#InstanceDetails:instanceId=i-0a84a91aa71762b54

EC2 > Instances > i-0a84a91aa71762b54

Instance summary for i-0a84a91aa71762b54 (panins14) [Info](#)

Updated less than a minute ago

Instance ID i-0a84a91aa71762b54	Public IPv4 address 52.66.119.69 open address	Private IPv4 addresses 172.31.15.41
IPv6 address -	Instance state Running	Public IPv4 DNS ec2-52-66-119-69.ap-south-1.compute.amazonaws.com open address
Hostname type IP name: ip-172-31-15-41.ap-south-1.compute.internal	Private IP DNS name (IPv4 only) ip-172-31-15-41.ap-south-1.compute.internal	Elastic IP addresses -
Answer private resource DNS name IPv4 (A)	Instance type t2.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendation Learn more
Auto-assigned IP address 52.66.119.69 [Public IP]	VPC ID vpc-055d721b1004494c1	Auto Scaling Group name -
IAM Role -	Subnet ID subnet-067810c72322625a8	Managed -
IMDSv2 -	Instance ARN -	

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Now again save the Public IPv4 address in the text file and compare it with the previous IP. It is evident that the IP address changes every time the instance is launched. To avoid it Elastic IP has to be assigned.

STEP 4-> From the EC2 Dashboard, go to the "Elastic Ips" option.

Elastic IP addresses | EC2 | ap-south-1 | pxeh2oy622al5wsw7konnjqu | ydr567igtbnxig6ycjion5ja0uc | +

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#Addresses:

Elastic Block Store > Elastic IPs > Elastic IP addresses

Elastic IP addresses

[Find elastic IP addresses by attribute or tag](#)

Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record
No Elastic IP addresses found in this Region				

[View IP address usage and recommendations to release unused IPs with Public IP insights.](#)

Allocate Elastic IP address

https://ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#Addresses: © 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences 32°C Haze 7:36 PM 4/25/2025

STEP 5-> Click on "Allocate Elastic IP" button on the top right corner of the screen. STEP 8-> With "Amazon's pool of IPv4 Addresses" selected under Public IPv4 Address pool, click on "Allocate".

Allocate Elastic IP address | EC2 x pxeh2oy622al5wws7konnjqu x ydr567igjtbngxig6ycjion5jaouc x +

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#AllocateAddress:

EC2 > Elastic IP addresses > Allocate Elastic IP address

Allocate Elastic IP address [Info](#)

Elastic IP address settings [Info](#)

Public IPv4 address pool

- ☒ Amazon's pool of IPv4 addresses
- ☐ Public IPv4 address that you bring to your AWS account with BYOIP. (option disabled because no pools found) [Learn more](#)
- ☐ Customer-owned pool of IPv4 addresses created from your on-premises network for use with an Outpost. (option disabled because no customer owned pools found) [Learn more](#)
- ☐ Allocate using an IPv4 IPAM pool (option disabled because no public IPv4 IPAM pools with AWS service as EC2 were found)

Network border group [Info](#)

ap-south-1

Global static IP addresses

AWS Global Accelerator can provide global static IP addresses that are announced worldwide using anycast from AWS edge locations. This can help improve the availability and latency for your user traffic by using the Amazon global network. [Learn more](#)

[Create accelerator](#)

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

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Search

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Allocate Elastic IP address | EC2 x pxeh2oy622al5wws7konnjqu x ydr567igjtbngxig6ycjion5jaouc x +

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#AllocateAddress:

EC2 > Elastic IP addresses > Allocate Elastic IP address

customer owned pools found) [Learn more](#)

- ☐ Allocate using an IPv4 IPAM pool (option disabled because no public IPv4 IPAM pools with AWS service as EC2 were found)

Network border group [Info](#)

ap-south-1

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AWS Global Accelerator can provide global static IP addresses that are announced worldwide using anycast from AWS edge locations. This can help improve the availability and latency for your user traffic by using the Amazon global network. [Learn more](#)

[Create accelerator](#)

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tag

[Cancel](#) [Allocate](#)

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Search

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STEP 6-> The Elastic IP has been created. Click on the IP to open it. Then Click on "Associate Elastic IP Address".

aws Elastic IP addresses | EC2 | ap-south-1 console.aws.amazon.com/ec2/home?region=ap-south-1#Addresses:PublicIp=13.202.154.202

EC2

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 - AMI Catalog
- Elastic Block Store
 - Volumes
 - Snapshots

Elastic IP address allocated successfully.
Elastic IP address 13.202.154.202

Associate this Elastic IP address

Elastic IP addresses (1)

Find elastic IP addresses by attribute or tag

Public IPv4 address: 13.202.154.202

	Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record
☐	-	13.202.154.202	Public IP	eipalloc-0c00fb5064313ec91	-

View IP address usage and recommendations to release unused IPs with [Public IP insights](#)

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STEP 7-> With resource type selected as "Instance", select the name of the instance & its private IP address. Check the bottom checkbox to reassociate the Elastic IP. Then click on "Associate".

Associate Elastic IP address | EC2 | ap-south-1 console.aws.amazon.com/ec2/home?region=ap-south-1#AssociateAddressDetails:PublicIp=13.202.154.202;allocationId=eipalloc-0c00fb5064313ec91

EC2 > Elastic IP addresses > 13.202.154.202 > Associate Elastic IP address

Elastic IP address: 13.202.154.202

Resource type

Choose the type of resource with which to associate the Elastic IP address.

☒ Instance

☐ Network interface

If you associate an Elastic IP address with an instance that already has an Elastic IP address associated, the previously associated Elastic IP address will be disassociated, but the address will still be allocated to your account. [Learn more](#)

If no private IP address is specified, the Elastic IP address will be associated with the primary private IP address.

Instance

Private IP address

The private IP address with which to associate the Elastic IP address.

Reassociation

Specify whether the Elastic IP address can be reassociated with a different resource if it already associated with a resource.

☒ Allow this Elastic IP address to be reassociated

Cancel Associate

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Elastic IP address | EC2 | ap-sou-1

pxeh2oy622al5wws7konnjqu... ydr567gjtbnxig6yjcjon5ja0uc...

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#ElasticIpDetails:PublicIp=13.202.154.202

EC2 > Elastic IP addresses > 13.202.154.202

EC2

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- Elastic Block Store**
 - Volumes
 - Snapshots

Elastic IP address associated successfully.
Elastic IP address 13.202.154.202 has been associated with instance i-0a84a91aa71762b54

13.202.154.202

Summary

Allocated IPv4 address 13.202.154.202	Type Public IP	Allocation ID eipalloc-0c00fb5064313ec91	Reverse DNS record -
Association ID eipassoc-06f45df612c10bf8a	Scope VPC	Associated instance ID i-0a84a91aa71762b54	Private IP address 172.31.15.41
Network interface ID eni-03f025982ae31cba9	Network interface owner account ID 676206917408	Public DNS ec2-13-202-154-202.ap-south-1.compute.amazonaws.com	NAT Gateway ID -
Address pool Amazon	Network border group ap-south-1		

Tags(0)

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STEP 8-> Copy the IPv4 address before selecting "Stop Instance" and again start this instance and copy the IPv4 address. STEP 13-> No changes are observed.

Instance details | EC2 | ap-south-1

pxeh2oy622al5wws7konnjqu... ydr567gjtbnxig6yjcjon5ja0uc...

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#InstanceDetails:instanceId=i-0a84a91aa71762b54

EC2 > Instances > i-0a84a91aa71762b54

EC2

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- Elastic Block Store**
 - Volumes
 - Snapshots

Instance summary for i-0a84a91aa71762b54

Updated less than a minute ago

Instance ID
i-0a84a91aa71762b54

IPv6 address
-

Hostname type
IP name: ip-172-31-15-41.ap-south-1.compute.internal

Answer private resource DNS name
IPv4 (A)

Auto-assigned IP address
-

IAM Role
-

IMDSv2
-

Public IPv4 address copied
13.202.154.202 | open address

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-15-41.ap-south-1.compute.internal

Instance type
t2.micro

VPC ID
vpc-055d721b1004494c1

Subnet ID
subnet-067810c72322625a8

Instance ARN
-

Private IPv4 addresses
172.31.15.41

Public IPv4 DNS
ec2-13-202-154-202.ap-south-1.compute.amazonaws.com | open address

Elastic IP addresses
13.202.154.202 [Public IP]

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Auto Scaling Group name
-

Managed
-

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Instance details | EC2 | ap-south-1

pxeh2oy622al5wws7konnjqu... ydr567gjtbnxig6yjcjon5ja0uc...

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#InstanceDetails:instanceId=i-0a84a91aa71762b54

EC2

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Successfully initiated stopping of i-0a84a91aa71762b54

Instance summary for i-0a84a91aa71762b54 (panins14)

Updated less than a minute ago

Instance ID
i-0a84a91aa71762b54

Public IPv4 address
13.202.154.202 | open address

Private IPv4 addresses
172.31.15.41

Instance state
Stopped

Public IPv4 DNS
ec2-13-202-154-202.ap-south-1.compute.amazonaws.com | open address

IPv6 address
-

Private IP DNS name (IPv4 only)
ip-172-31-15-41.ap-south-1.compute.internal

Elastic IP addresses
13.202.154.202 [Public IP]

Hostname type
IP name: ip-172-31-15-41.ap-south-1.compute.internal

Instance type
t2.micro

Answer private resource DNS name
IPv4 (A)

VPC ID
vpc-055d721b1004494c1

Auto-assigned IP address
-

AWS Compute Optimizer finding
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IAM Role

Subnet ID
subnet-06381073232625f812

Auto Scaling Group name

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Instance details | EC2 | ap-south-1

pxeh2oy622al5wws7konnjqu... ydr567gjtbnxig6yjcjon5ja0uc...

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#InstanceDetails:instanceId=i-0a84a91aa71762b54

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Snapshots

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Successfully initiated starting of i-0a84a91aa71762b54

Notifications 0 0 2 0 0 0

Instance summary for i-0a84a91aa71762b54 (panins14)

Updated less than a minute ago

Instance ID
i-0a84a91aa71762b54

Public IPv4 address
13.202.154.202 | open address

Private IPv4 addresses
172.31.15.41

Instance state
Running

Public IPv4 DNS
ec2-13-202-154-202.ap-south-1.compute.amazonaws.com | open address

IPv6 address
-

Private IP DNS name (IPv4 only)
ip-172-31-15-41.ap-south-1.compute.internal

Elastic IP addresses
13.202.154.202 [Public IP]

Hostname type
IP name: ip-172-31-15-41.ap-south-1.compute.internal

Instance type
t2.micro

Answer private resource DNS name
IPv4 (A)

VPC ID
vpc-055d721b1004494c1

Auto-assigned IP address
-

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IAM Role

Subnet ID

Auto Scaling Group name

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NOW observed that:

```
Before Assigning Elastic IP:
3.7.65.189
52.66.119.69
After Assigning Elastic IP:
13.202.154.202
13.202.154.202
```


